Syllabus

This syllabus is subject to change. You will be notified of any changes as far in advance as possible via an announcement on Brightspace. Monitor your Purdue email daily for updates.

Course Information

Course number and title: AAE 20401 Aeromechanics II Laboratory

Instructional modality: In-person labs and asynchronous online lectures

CRNs, meeting times, and locations:

Section	CRN	Days	Time	Location
001	10003	T	1:30-3:20 pm	ARMS 3106
002	10004	T	9:30-11:20 am	ARMS 3106
006	10008	W	9:30-11:20 am	ARMS 3106
009	10011	M	1:30-3:20 pm	ARMS 3106
011	37250	T	3:30-5:20 pm	ARMS 3106
013	43648	M	3:30-5:20 pm	ARMS 3106
015	47686	M	11:30-1:20 pm	ARMS 3106
017	18954	M	9:30-11:20 am	ARMS 3106
018	18957	T	11:30-1:20 pm	ARMS 3106
019	23081	W	11:30-1:20 pm	ARMS 3106
020	10020	-	Asynchronous online	-
021	26753	W	1:30-3:20 pm	ARMS 3106
022	26755	W	3:30-5:20 pm	ARMS 3106
023	26756	R	9:30-11:20 am	ARMS 3106
024	16227	R	11:30-1:20 pm	ARMS 3106
025	16229	R	1:30-3:20 pm	ARMS 3106
026	16230	R	3:30-5:20 pm	ARMS 3106
027	29981	F	11:30-1:20 pm	ARMS 3106
028	29982	F	9:30-11:20 am	ARMS 3106
029	39023	F	1:30-3:20 pm	ARMS 3106

Prerequisites: None

Co-requisites: AAE 20400 Aeromechanics II

Credits: 1.0

Learning Resources, Technology & Texts

Required textbook: No textbook is required.

Brightspace learning management system (LMS): All course materials including lab manuals will be uploaded to Purdue's Brightspace learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. It is strongly suggested that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Gradescope: Your assignments will be submitted to Gradescope for grading.

Course Description

This course is to introduce aerospace engineering students to strain gauges and Wheatstone bridge, as well as mechanics of materials using tensile testing, basic truss structures, bending of I-beams, torsion of shafts, and bi-axial stress and strain.

Learning Outcomes

On completing this course, the student shall be able to:

1. Master the basic methods in the experimental testing of solids and structures.
2. Understand and set up experimental instrumentation.
3. Set up and conduct mechanical tests on materials in tension and shear.
4. Understand the concept of stress concentration in structures.
5. Compute stresses, strains, deformation and displacements of basic structural components.
6. Set up and conduct beam bending experiments.
7. Master the basic skills in the writing of technical reports.

Course Goals and Objectives

The objective of this course is to provide the student with the foundation of mechanical testing measurements through guided experiments. By understanding material behavior, modes of loading, stress–strain relation, and effects of multi-axial stresses, students will be able to better evaluate the conditions for meeting design requirements in aerospace engineering. The key elements in this course are:

- **Instrumentation:** Understanding of potential and limitation of experimental measurements. Setup, calibrate, and read from sensors. Be able to plan an experiment and decide what sensor to use.
- **Data manipulation:** Fitting equations, data manipulation, determination of trends and understanding of data scatter.
- **Material behavior:** Understanding of elastic and plastic deformation, stress–strain behavior, Poisson's ratio, effects of stress concentration, shaft torsion, beam bending, and biaxial stresses.

Assignments and Grading

Assignments	Due	Weight
Pre-lab assignments	Before each lab	20%
Lab reports	One week after each lab	80%

Pre-lab Assignments

- Pre-lab assignments are to be completed individually. The pre-lab assignments will contain

questions related to the pre-lab lecture material and the virtual labs.
- The lab instructor will decide on the logistics of the pre-lab assignment submission.
- No late assignments will be accepted.

Lab Reports
- Lab reports are to be completed as a group. Lab experiments will also be completed as a group and experimental data may be shared among the group members. The groups will be randomly selected by the lab instructor during each lab session.
- Reports are to be typed with the student names, section, and lab instructor indicated on the cover page.
- You may not use data collected by other lab groups/sections unless you have permission from the lab instructor. Copying another group's report or using another group's experimental data will automatically result in a letter grade of F.
- Lab reports must be submitted one week after the experiment is completed. The lab instructor will decide the logistics of the report submission.
- No late lab reports will be accepted.
- A dedicated ***Authorship Contribution Statement*** must be included in each lab report to clearly document the specific contributions of each team member. A template of Authorship Contribution Statement will be provided in the lab report template.

Peer Review
- After each hands-on lab experiment, every student will individually do a peer review of the performance of each of their lab/group mates.
- The peer review form will be made available on Brightspace and must be submitted individually one week after the hands-on lab experiment.
- Completing a peer review will not contribute towards your final grade. However, if you receive multiple negative peer reviews from your lab/group mates, your grade may be scaled down at the lab instructor's discretion.

Responsible Use of Artificial Intelligence (AI) Tools

The responsible use of artificial intelligence (AI) tools (such as ChatGPT, Copilot, Gemini, etc.) can enhance learning and creativity when applied appropriately. In this course, AI tools may be used as a ***supplement to learning***, but not as a replacement for your own critical thinking and original work. All submitted work must reflect your own understanding. Misuse of AI tools will be considered a violation of Academic Integrity.

Prelab
- AI tools may be used to explore concepts, clarify background knowledge, or provide practice examples.
- Students should **not** submit any AI-generated responses. All solutions, derivations, and explanations must be original and based on their understanding.

Lab Report
- AI may be used for spell-check and grammar suggestions.
- **AI-generated text, figures, or code may not be directly included in the lab report.** All writing, analysis, and presentation materials must be authored by the student(s).
- If AI tools are used for idea generation, students should acknowledge this in their project documentation (e.g., "AI tools were consulted to explore brainstorming ideas for initial design concepts").

Grading Scale

Your grade is based on your performance in the pre-lab assignments and lab reports. Additionally, your grades may be scaled based on the feedback you receive from your colleagues in your peer review. At the end of the semester, final grades will be calculated by adding the total points earned and translating those percentages into the following letters.

A+ ≥ 96.50%
96.49% ≥ A ≥ 93.50%
93.49% ≥ A- ≥ 89.50%

89.49% ≥ B+ ≥ 86.50%
86.49% ≥ B ≥ 83.50%
83.49% ≥ B- ≥ 79.50%

79.49% ≥ C+ ≥ 76.50%
76.49% ≥ C ≥ 73.50%
73.49 ≥ C- ≥ 69.50%

69.49% ≥ D+ ≥ 66.50%
66.49% ≥ D ≥ 63.50%
63.49% ≥ D- ≥ 59.50%

F ≤ 59.49%

The instructor reserves the right to curve or to not curve the class.

Virtual Lab Activities

For each hands-on lab activity and pre-lab assignment, you will need to complete a virtual lab activity.

- The virtual lab activities are designed to replicate the actual hands-on lab activities and to give you some familiarity with the experimental equipment.
- They will also allow you to perform 'virtual' experiments which you may not be able to do using the actual experimental setup.
- In order to complete the virtual lab activity, you will need to use the virtual lab software Virtual Lab 0.67b.
- You have two options for using this software:
    - Option 1: Use one of the computers in ARMS 2106. Each computer has the virtual lab software installed.
    - Option 2: Use AppAnywhere to remotely access the virtual lab software. AppsAnywhere is a program that allows you to use any of the software available on the ECN computers on your own laptop/home computer. The virtual lab software is available on AppsAnywhere. The following link describes how to use AppsAnywhere: https://engineering.purdue.edu/ECN/Support/KB/Docs/UsingAppsAnywhere.1
- Once you have access to the software (using either option 1 or 2 above), please consult the virtual lab manual uploaded to Brightspace. This document will help you navigate step-by-step each of the virtual lab activities.
- **The pre-lab assignments will contain questions related to the virtual lab activities**.

Attendance Policy

The lectures will be posted online to Brightspace every other week. You are expected to watch the lectures in your own time and complete the pre-lab assignment before attending the in-person lab activity. **Attending the lab activities is mandatory. If you miss a lab activity without a pre-approved excuse, you will automatically receive a F grade for that lab with no chance of repeating the lab activity.**

Course Schedule

Holidays are indicated in **bold blue**.

Week	Dates	Lab Preparation Lecture	Lab Session	Assignments
1	1/12 – 17	**Course Introduction**	No Lab	
2	1/19 – 1/24	**MLK Day on 1/20** **Lecture 1** • Strain gages • Wheatstone bridge • Pre-lab assignment	**In-person lab orientation in ARMS 3106** (Students in the Monday sessions please follow your lab TA's instruction.)	Pre-lab 1 due on **1/23**
3	1/27 – 1/31	No lecture	**Lab 1 – Strain gages and Wheatstone bridge** • Measuring strain gage resistance using multi-meter • Measuring strain using a Wheatstone bridge	
4	2/2 – 2/7	**Lecture 2** • Linear elasticity • Material behavior • Tensile tests • Pre-lab assignment	No lab	Lab report 1 due Pre-lab 2 due on 2/6
5	2/9 – 2/14	No lecture	**Lab 2 – Material characterization** • Tensile testing of aluminum bar • Obtain axial stress versus strain response • Obtain transverse strain versus axial strain response • Calculate Young's modulus and Poisson's ratio	
6	2/16 – 2/21	**Lecture 3** • Introduction to trusses • 3D trusses • Analysis of truss structures using method of joins, method of sections • Truss elements • Example problem • Pre-lab assignment	No lab	Lab report 2 due Pre-lab 3 due on 2/20
7	2/23 – 2/28	No lecture	**Lab 3 – Trusses: Basic structures**	

			• Find internal loads in truss structure using axial strain measurement • Compare experimental and theoretical results	
8	3/2 – 3/7	**Lecture 4** • Beams in aerospace structures • Shear force and bending moment diagrams • Cross-sectional properties • Small deflection of beams • Pre-lab assignment	No lab	Lab report 3 due Pre-lab 4 due on 3/6
9	3/9 – 3/14	No lecture	**Lab 4 – Beam bending** • Four-point bending an I-beam • Obtain shear force and bending moment diagrams • Compare experimental and theoretical results	
10		**Spring Break (3/16 – 3/21)** No lectures or labs		
11	3/23 – 3/28	**Lecture 5** • Shear and torsion in aerospace structures • Deformation due to torsion • Stress-strain behavior in torsion • Simple shear loading • Pre-lab assignment	No lab	Lab report 4 due Pre-lab 5 due on 3/27
12	3/30 – 4/4	No lecture	**Lab 5 – Torsion and shear** • Torsion of a shaft • Measure angle of twist and shear strain • Plate under simple shear • Plot shear stress-shear strain behavior • Calculate shear modulus	
13	4/6 – 4/11	**Lecture 6** • Pressure vessels in aerospace structures	No lab	Lab report 5 due

		• Hooke's law for multi-axial stress states • Strain transformation • Thin-walled pressure vessels • Virtual lab • Pre-lab assignment		Pre-lab 6 due on 4/10
14	4/13 – 4/18	No lecture	**Lab 6 – Biaxial stress and strain** • Uniaxial tensile testing • Plot strain versus applied load • Thin-walled pressure vessel • Plot strain versus applied pressure • Compare experimental and theoretical results	
15	4/20 – 4/25	No lecture	No lab	Lab report 6 due
16	4/27 – 5/2	No lecture	No lab	
17	5/4 – 5/9	Final exam (no lectures or labs)		

** Schedule and assignments subject to change. Any changes will be posted in the learning management system.*

Course Evaluation

During the last two weeks of the semester, you will be provided with an opportunity to give feedback on this course and your instructor. Purdue uses an online course evaluation system. You will receive an official email from evaluation administrators with a link to the online evaluation site. You will have up to 10 days to complete this evaluation. Your participation is an integral part of this course, and your feedback is vital to improving education at Purdue University. I strongly urge you to participate in the evaluation system.

Netiquette

Your instructor and fellow students wish to foster a safe online learning environment. All opinions and experiences, no matter how different or controversial they may be perceived, must be respected in the tolerant spirit of academic discourse. You are encouraged to comment, question, or critique an idea, but you are not to attack an individual. Our differences, some of which are outlined in the University's nondiscrimination policy statement, will add richness to this learning experience. Please consider that sarcasm and humor can be misconstrued in online interactions and generate unintended disruptions. Working as a community of learners, we can build a polite and respectful course ambience. Please read the Netiquette rules for this course:

- Monitor how much space/time you are taking up in any discussion. Give other students the opportunity to join in the discussion.
- Do not use offensive language. Present ideas appropriately.
- Be cautious in using Internet language. For example, do not capitalize all letters since this suggests shouting.
- Avoid using vernacular and/or slang language. This could lead to misinterpretation.
- Keep an "open-mind" and be willing to express even your minority opinion.

- Think and edit before you push the 'Send' button.

Seek and take in feedback from others; learning from other people is an important life skill.

Diversity & Inclusion Statement

We strive for equity, providing equal access and opportunity, and working to maximize student potential. This requires both instructor and students to identify and remove barriers that may prevent someone from full access or full participation. You can help by:

- Contacting me, anonymously if needed, if you see a potential barrier for someone or yourself in participating fully in the class. This might be a physical barrier such as access to technology or a personal situation.
- Suggesting ways in which members of our class can support each other. Virtual study groups and discussion boards are examples, but I encourage you to be creative in your ideas.

Getting to know each other as contributing members of our learning community. Everyone has something to contribute, and while I designed the course to take advantage of the wealth of knowledge, expertise, and experience we bring together, I cannot do it well without your participation. There are many opportunities built into this course for this type of work. It is important we do it together.

Emergency Preparation

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

A link to Purdue's Information on Emergency Preparation and Planning is located on our Brightspace under "University Policies and Statements." This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System. I encourage you to review the *Emergency Preparedness for Classrooms document* attached in this syllabus.

Please read the resources under **University Policies and Statements** and the **Student Services** widget in Brightspace regarding **Technology Resources**, **Academic Help**, **Campus Resources**, **Academic Integrity**, **Nondiscrimination Policy Statement**, **Copyright**, and **Emergency Preparedness**.



EMERGENCY PREPAREDNESS for Classrooms

EMERGENCY NOTIFICATION PROCEDURES are based on a simple concept – if you hear a fire alarm inside, proceed outside. If you hear a siren outside, proceed inside.

- **Indoor Fire Alarms** mean to stop class or research and immediately **evacuate** the building.
  - Proceed to your Emergency Assembly Area away from building doors. **Remain outside** until police, fire, or other emergency response personnel provide additional guidance or tell you it is safe to leave.

- **All Hazards Outdoor Emergency Warning Sirens** mean to immediately seek shelter (**Shelter in Place**) in a safe location within the closest building.
  - "Shelter in place" means seeking immediate shelter inside a building or University residence. This course of action may need to be taken during a tornado, an active threat such as a shooting or release of hazardous materials in the outside air. Once safely inside, find out more details about the emergency*. **Remain in place** until police, fire, or other emergency response personnel provide additional guidance or tell you it is safe to leave.

**In both cases, you should seek additional clarifying information by all means possible…Campus Emergency Status page, text message, email alert, TV, radio, etc…review the Purdue Emergency Warning Notification System multi-communication layers at http://www.purdue.edu/ehps/emergency_preparedness/warning-system.html*

EMERGENCY RESPONSE PROCEDURES:

- Review the **Emergency Procedures Guidelines** https://www.purdue.edu/emergency_preparedness/flipchart/index.html
- Review the **Building Emergency Plan** (available on the Emergency Preparedness website or from the building deputy) for:
  - evacuation routes, exit points, and emergency assembly area
  - when and how to evacuate the building.
  - shelter in place procedures and locations
  - additional building specific procedures and requirements.

EMERGENCY PREPAREDNESS AWARENESS VIDEO

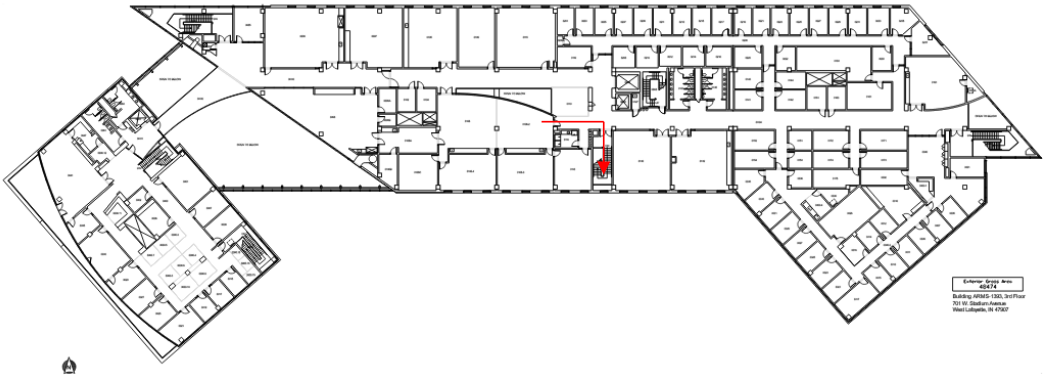
- **"Run. Hide. Fight.®"** is a 6-minute active shooter awareness video that illustrates what to look for and how to prepare and react to this type of incident. See: https://www.youtube.com/watch?v=5mzI_5aj4Vs

MORE INFORMATION
Reference the Emergency Preparedness web site for additional information:
http://www.purdue.edu/ehps/emergency_preparedness

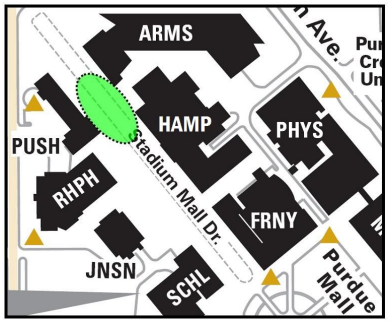
As of July 1, 2024

ARMS 3106

Fire Alarm:



If we hear a fire alarm, we will immediately evacuate the building and proceed to Stadium Mall between PUSH and HAMP. In case of inclement weather, meet in Hampton Hall (HAMP).



All Hazards Warning Sirens:

If we are notified of a Shelter in Place requirement for a tornado warning, we will shelter in the lowest level of this building away from windows and doors. Our preferred locations are in lecture halls B061, B071, and 1010.